# Dynamic Causal Attack Graph Construction from Unstructured Cyber Threat Intelligence Reports

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# Problem Statement

Cyber Threat Intelligence (CTI) reports contain critical information about attackers, malware, vulnerabilities but this information is mostly unstructured text, making it difficult to automatically extract and utilize for security analysis. Existing approaches using general-purpose NLP models often fail to accurately capture domain-specific cybersecurity entities and relationships, limiting their effectiveness in constructing attack graphs and predicting future cyber attack events.

# Objectives

- To develop an end-to-end framework to extract knowledge from CTI reports, construct attack graphs, and forecast future cyber attack events.
- To create a domain-adapted BERT model.
- To construct Attack Graph and do Attack Forecasting for predicting future cyber attack events.

# Literature Survey

| Paper Title | Year | Journal | Methodology | Advantages | Disadvantages |
|---|---|---|---|---|---|
| DcBERT-IC: Framework for Cyber Threat Analysis Integrating Deep Learning Model and Attack Intelligence Graph | 2025 | IEEE Transactions on Consumer Electronics | Uses a DcBERT-BiGRU-CRF model for NER and DcBERT-SVD-LightGBM for attack-cycle classification. Detected entities are mapped to MITRE ATT&CK and visualized as Attack Intelligence Graphs. | Very high accuracy (~98.40%) and recall (~94.9%). Integrates NER, classification, and visualization. More efficient compared to typical DL pipelines. | Requires large training datasets, high computational cost. Performance depends on CTI writing quality and may not operate fully in real-time. |
| MTRetrieval: Retrieving MITRE Technique from Unstructured CTI Reports by Fusing Deep Learning and Ontology | 2024 | IEEE Transactions on Network and Service Management | Uses Sentence-level BERT (SBERT) combined with COMAT ontology and level-wise weighting to classify implicit MITRE ATT&CK techniques from CTI reports. | Handles implicit TTP detection. Ontology improves semantic understanding. Achieves ~90% accuracy even with few training samples. | Requires ontology integration, higher complexity. Processing long reports (3–10 min) per sample. Manual review sometimes needed. |

# Literature Survey

| Paper Title | Year | Journal | Methodology | Advantages | Disadvantages |
|---|---|---|---|---|---|
| Nip in the Bud: Forecasting Emerging Post-Exploitation Attacks in Real-Time Through CTI Reports | 2025 | IEEE Transactions on Dependable and Secure Computing | Extracts Attack Scene Graphs (ASG) from CTI. Trains forecast models combining EDR Attack Provenance Graph (APG) to predict future Attack Graph (AFG) patterns using Attack Temporal Graphs (ATG). | Enables real-time post-exploitation forecasting. Strong CTI system integration. Achieves ~91.8% prediction accuracy with reduced false positives. | Requires EDR integration and complex graph pipelines. Sensitive to ASG extraction quality. |
| TRACE: Relationship Analysis and Causal Factor Extraction for Cyber Threat Intelligence Reports | 2025 | IEEE Transactions on Dependable and Secure Computing | Uses SBERT + BiLSTM knowledge mapping to extract causal factors (CWE + Data Components) and generate causal dependency graphs linking ATT&CK techniques. | Introduces causal reasoning linking attacks to vulnerabilities. Improves prediction accuracy (~87%). Helps in mitigation planning. | Focused mainly on causal relationships, not full attack sequences. Relies heavily on accurate CWE mapping. Limited benchmarking datasets. |

# Summary

**Research Gap Identified**

- Use general language models not optimized for cybersecurity text.
- Use standard CRF that captures only local label dependencies.

**Proposed Solution**

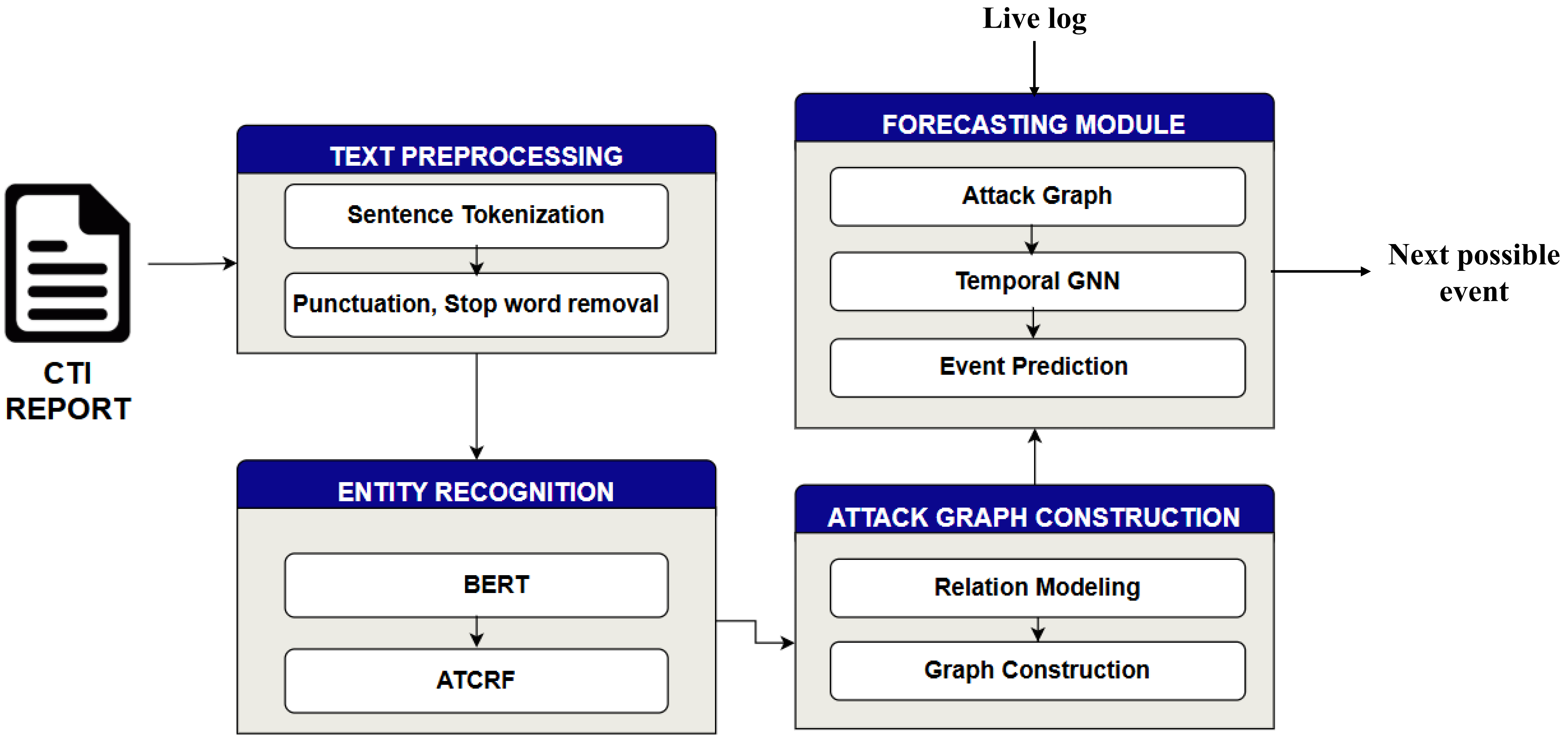
- Domain Adaptive Pretraining (DAPT) on CTI reports
- Context-aware CRF decoding using attention-based transition scoring
- Improved CTI entity extraction accuracy

# Dataset Description

- Domain Adapted BERT
  - A collection of threat reports, blogs
  - Corpus size : 95 MB
  - No. of reports – 10,000
- NER
  - Harmonized STIX Dataset – BIO tagged
  - No of tokens ~ 6 lakh
  - 24 entities considered

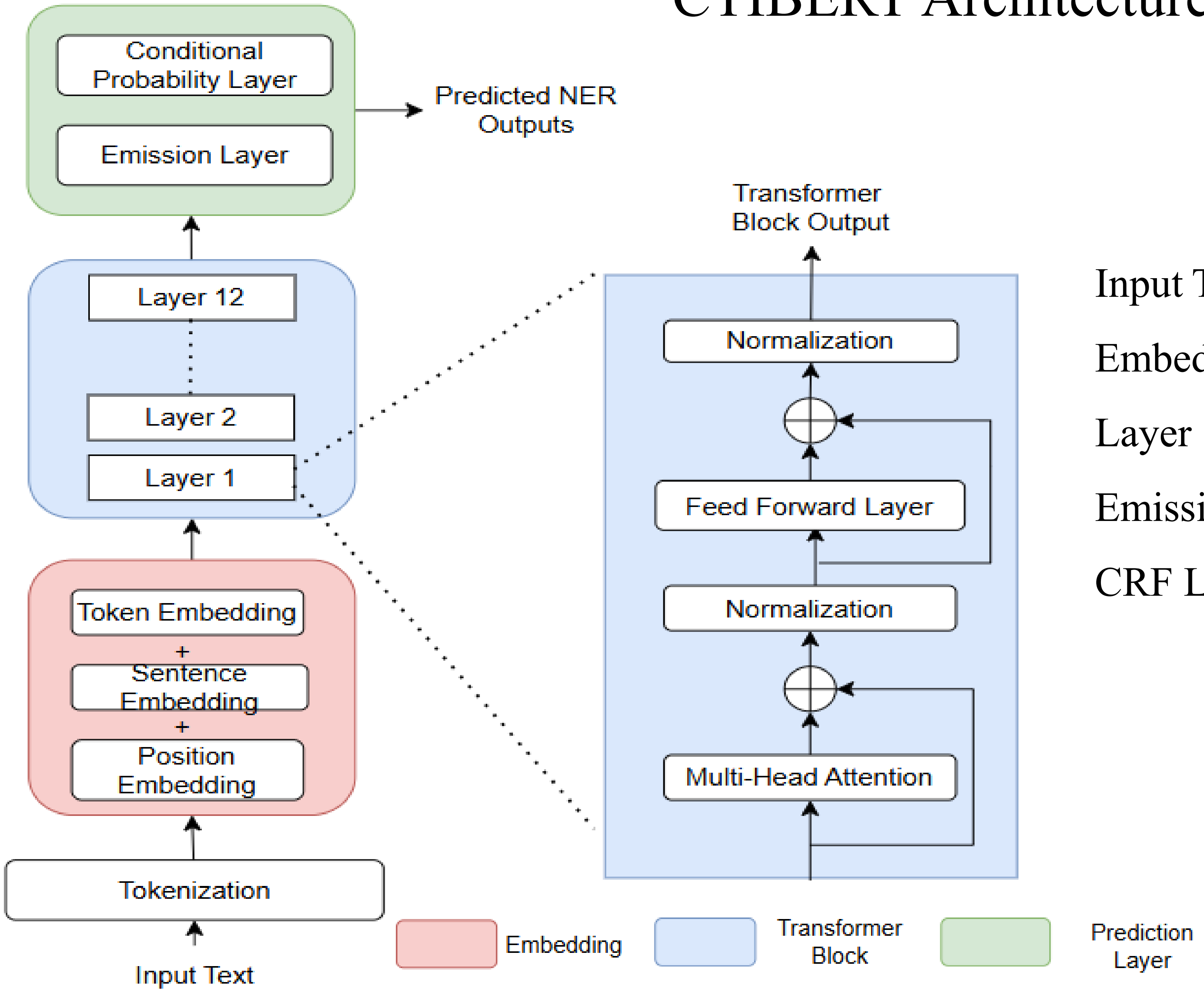
| **Entity Type** |
|---|
| Vulnerability |
| Exploit |
| Threat-Actor-Group |
| Threat-Actor-Organization |
| Malware-Ransomware |
| Malware-Trojan |
| Malware-Worm |
| Attack-Initial-Access |
| Attack-Execution |
| Attack-Persistence |
| Attack-Privilege-Escalation |

| **Entity Type** |
|---|
| Indicator-IP |
| Indicator-Domain |
| Indicator-URL |
| Indicator-File-Hash |
| Tool |
| File-Name |
| Process-Name |
| Location-Country |
| Location-City |
| Time-Date |
| Time-Duration |
| Campaign |
| Organization |




Architecture Diagram

# CTIBERT Architecture




Input Tokens → (N)

Embedding Layer → (N, 768)

Layer 1 - 12 → (N, 768)

Emission Layer → (N, 24)

CRF Layer → (N)

# Entity Recognition

**Input: CTI sentences X, BIO labels Y**
**Output: Trained CTI-BERT**

```
Initialize BERT
Add emission layer
Initialize base transition matrix A
Add context transformation layer g(c)
For each sentence X:
    Tokenize sentence
    Pass tokens through CTI-BERT
    Obtain contextual embeddings H
    Compute emission scores: E = W * H + b
    Compute sentence context vector: c = mean(H)
    Compute adaptive transition: A' = A + g(c)
    Compute CRF score: Score(X,Y)
    Compute Joint probability
    Update parameters using backpropagation
During inference:
    Compute emission scores
    Compute adaptive transition matrix A’
    Apply Viterbi decoding using A’
Return predicted labels
```

Standard CRF - Existing

$$Score\ (X,Y) = \sum A + \mathrm{E}$$

Adaptive Transition CRF - Proposed

$$Score\ (X,Y) = \sum \mathrm{A}' + \mathrm{E}$$

$$A' = A + g(c)$$

$$P(Y \mid X) = \frac{\exp(\mathrm{Score}(X,Y))}{\sum_{Y'} \exp(\mathrm{Score}(X,Y'))}$$

# Forecasting

**Forecasting**

**Input : Graph dataset $\mathcal{G}$**

**Output: future event**

```
for each graph G_k in 𝓖 do
    Extract node feature matrix X from G_k
    // GNN Layers
    Z^1 = σ(A · X · W_1)          // Layer 1 aggregation
    Z^2 = σ(A · Z^1 · W_2)        // Layer 2 aggregation
    // Sequence Formation
    Arrange Z^2 as sequence:
    Z_seq = [z_1, z_2, ..., z_t]
    // Temporal Modeling
    H = Temporal_Model(Z_seq)   // LSTM
    h_t = final hidden state
    // Prediction
    y = Softmax(W · h_t)
    Predicted event:
    e_{t+1} = argmax(y)
end for
return predicted events
```

# Results

## Input Text

A threat actor known as APT29 launched a phishing campaign targeting Acme Corp and BlueRetail Ltd. The attackers sent emails containing a malicious invoice.pdf attachment that attempted to deliver the TrickBot malware. At the same time, another attacker group called FIN7 began scanning network infrastructure in Germany and France. The TrickBot malware attempted to exploit CVE-2021-34527 to gain initial access. Simultaneously, Emotet malware was delivered through a malicious payload.exe file. The attackers used PowerShell and Cobalt Strike as tools to establish persistence. Meanwhile, suspicious network traffic was detected communicating with an indicator 192.168.45.12. At the same time, another connection was observed to malicious-domain.xyz indicating command and control communication. The threat actor Lazarus Group also deployed Mirai malware targeting IoT devices. Concurrently, a phishing attack pattern attempted to steal credentials from employees of RetailBank Inc. The attackers used the tool Mimikatz to dump credentials from compromised machines. Meanwhile, malicious files such as credential_dump.txt were created on infected systems. At the same time, suspicious network traffic flows were detected between infected hosts and external servers. The malware Emotet attempted to propagate laterally using stolen credentials. Simultaneously, the threat actor APT28 launched another attack pattern exploiting CVE-2020-1472. The attackers used PsExec to move laterally within the corporate network. Meanwhile, additional malware DarkComet was installed on several machines. The attackers created files such as backdoor.dll to maintain persistence. Concurrently, network traffic indicators such as 185.23.44.90 were detected communicating with compromised hosts. Finally, the attackers exfiltrated sensitive data from Acme Corp and RetailBank Inc.

## Tokenization

```
Sentence 1:
A threat actor known as APT29 launched a phishing campaign targeting Acme Corp and BlueRetail Ltd.
Sentence 2: The attackers sent emails containing a malicious invoice.pdf attachment that attempted to deliver t
Sentence 3: At the same time, another attacker group called FIN7 began scanning network infrastructure in Germa
Sentence 4: The TrickBot malware attempted to exploit CVE-2021-34527 to gain initial access.
Sentence 5: Simultaneously, Emotet malware was delivered through a malicious payload.exe file.
Sentence 6: The attackers used PowerShell and Cobalt Strike as tools to establish persistence.
Sentence 7: Meanwhile, suspicious network traffic was detected communicating with an indicator 192.168.45.12.
Sentence 8: At the same time, another connection was observed to malicious-domain.xyz indicating command and co
Sentence 9: The threat actor Lazarus Group also deployed Mirai malware targeting IoT devices.
Sentence 10: Concurrently, a phishing attack pattern attempted to steal credentials from employees of RetailBan
Sentence 11: The attackers used the tool Mimikatz to dump credentials from compromised machines.
Sentence 12: Meanwhile, malicious files such as credential_dump.txt were created on infected systems.
Sentence 13: At the same time, suspicious network traffic flows were detected between infected hosts and extern
Sentence 14: The malware Emotet attempted to propagate laterally using stolen credentials.
Sentence 15: Simultaneously, the threat actor APT28 launched another attack pattern exploiting CVE-2020-1472.
Sentence 16: The attackers used PsExec to move laterally within the corporate network.
Sentence 17: Meanwhile, additional malware DarkComet was installed on several machines.
Sentence 18: The attackers created files such as backdoor.dll to maintain persistence.
Sentence 19: Concurrently, network traffic indicators such as 185.23.44.90 were detected communicating with com
Sentence 20: Finally, the attackers exfiltrated sensitive data from Acme Corp and RetailBank Inc.

Sentence 1 tokens:
['A', 'threat', 'actor', 'known', 'as', 'APT29', 'launched', 'a', 'phishing', 'campaign', 'targeting', 'A
Token count: 17

Sentence 2 tokens:
['The', 'attackers', 'sent', 'emails', 'containing', 'a', 'malicious', 'invoice.pdf', 'attachment', 'that
Token count: 17

Sentence 3 tokens:
['At', 'the', 'same', 'time', ',', 'another', 'attacker', 'group', 'called', 'FIN7', 'began', 'scanning',
Token count: 19

Sentence 4 tokens:
['The', 'TrickBot', 'malware', 'attempted', 'to', 'exploit', 'CVE-2021-34527', 'to', 'gain', 'initial', '
Token count: 12

Sentence 5 tokens:
['Simultaneously', ',', 'Emotet', 'malware', 'was', 'delivered', 'through', 'a', 'malicious', 'payload.ex
Token count: 12

Sentence 6 tokens:
['The', 'attackers', 'used', 'PowerShell', 'and', 'Cobalt', 'Strike', 'as', 'tools', 'to', 'establish', '
Token count: 13
```

| Sentence No. | Tokens in Original Sentence | After punct. removal | After Stopword Removal |
|---|---|---|---|
| 1 | 17 | 16 | 12 |
| 2 | 17 | 16 | 11 |
| 3 | 19 | 17 | 12 |
| 4 | 12 | 11 | 8 |
| 5 | 12 | 10 | 7 |
| 6 | 13 | 12 | 8 |
| 7 | 13 | 11 | 8 |
| 8 | 17 | 15 | 9 |
| 9 | 13 | 12 | 11 |
| 10 | 16 | 14 | 10 |

| Sentence No. | Tokens in Original Sentence | Count after punct. removal | Count after Stopword Removal |
|---|---|---|---|
| 11 | 13 | 12 | 8 |
| 12 | 13 | 11 | 7 |
| 13 | 18 | 16 | 10 |
| 14 | 11 | 10 | 8 |
| 15 | 13 | 11 | 10 |
| 16 | 12 | 11 | 8 |
| 17 | 11 | 9 | 7 |
| 18 | 11 | 10 | 6 |
| 19 | 15 | 13 | 9 |
| 20 | 14 | 12 | 9 |
| Total | 280 | 249 | 178 |

# CTI-BERT Layer wise summary

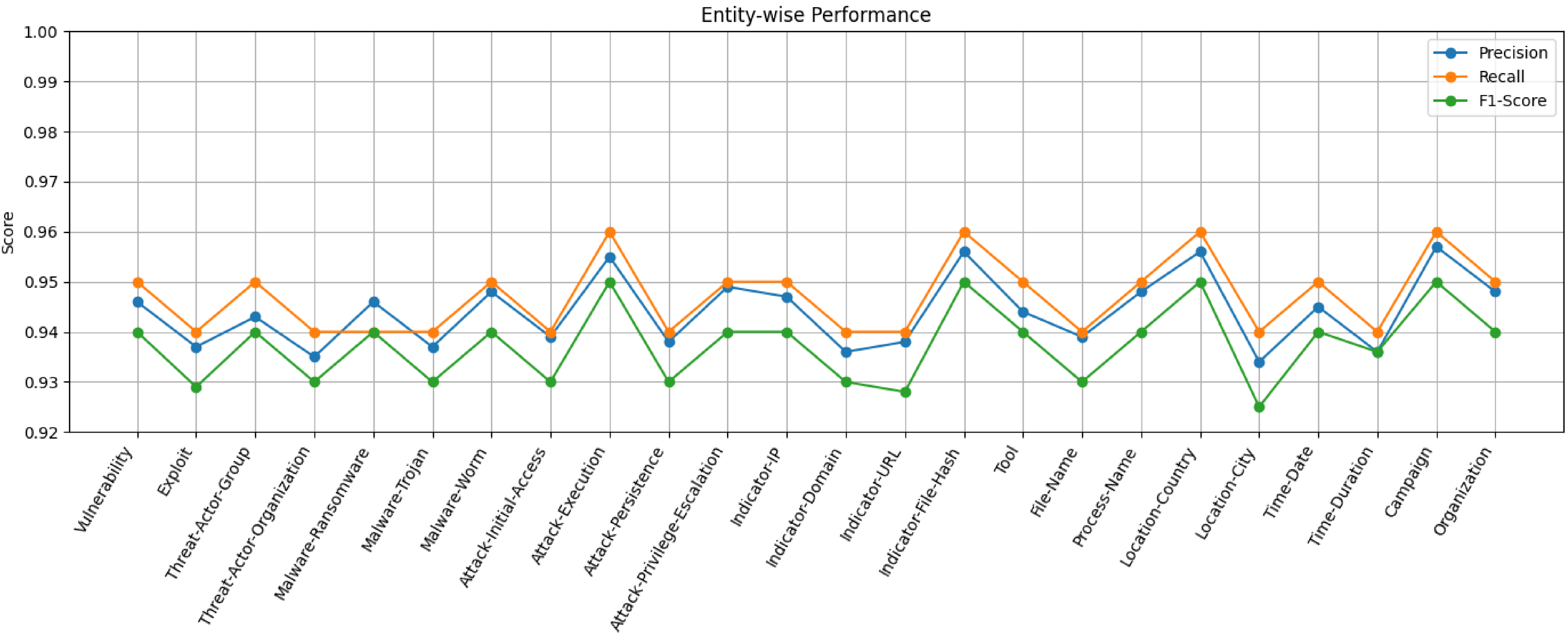
| Layer | Embedding | Similarity |
|---|---|---|
| 1 | apt28 → [-1.23, -1.02, -0.81, -0.79, -0.36]<br>executes → [-0.83, -1.54, -0.40, -0.72, -0.38]<br>powershell → [-1.00, -0.15, -0.85, -0.60, -0.75] | apt28 ↔ executes = 0.61<br>executes ↔ powershell = 0.64<br>apt28 ↔ powershell = 0.60 |
| 2 | apt28 → [-1.09, -0.28, -1.09, -0.12, -0.55]<br>executes → [-0.87, -0.63, -1.07, 0.07, -0.49]<br>powershell → [-0.90, 0.20, -1.12, 0.07, -0.54] | apt28 ↔ executes = 0.69<br>executes ↔ powershell = 0.71<br>apt28 ↔ powershell = 0.68 |
| 3 | apt28 → [-0.97, 0.30, -1.76, -0.35, 0.07]<br>executes → [-0.87, 0.04, -1.76, -0.22, 0.07]<br>powershell → [-0.86, 0.54, -1.79, -0.24, 0.24] | apt28 ↔ executes = 0.74<br>executes ↔ powershell = 0.78<br>apt28 ↔ powershell = 0.75 |
| 4 | apt28 → [-0.20, 0.06, -2.02, -0.43, 0.51]<br>executes → [-0.09, -0.03, -2.00, -0.31, 0.43]<br>powershell → [-0.18, 0.14, -2.00, -0.32, 0.59] | apt28 ↔ executes = 0.77<br>executes ↔ powershell = 0.81<br>apt28 ↔ powershell = 0.78 |

# CTI-BERT Layer wise summary

| Layer | Embedding | Similarity |
|---|---|---|
| 5 | apt28 → [-0.15, 0.20, -2.10, -0.30, 0.65]<br>executes → [-0.10, 0.18, -2.12, -0.28, 0.63]<br>powershell → [-0.12, 0.22, -2.15, -0.28, 0.70] | apt28 ↔ executes = 0.83<br>executes ↔ powershell = 0.87<br>apt28 ↔ powershell = 0.84 |
| 6 | apt28 → [-0.10, 0.30, -2.25, -0.25, 0.75]<br>executes → [-0.07, 0.28, -2.28, -0.23, 0.78]<br>powershell → [-0.08, 0.35, -2.30, -0.20, 0.80] | apt28 ↔ executes = 0.88<br>executes ↔ powershell = 0.91<br>apt28 ↔ powershell = 0.89 |
| 7 | apt28 → [-0.05, 0.40, -2.35, -0.20, 0.85]<br>executes → [-0.03, 0.38, -2.38, -0.19, 0.87]<br>powershell → [-0.03, 0.45, -2.40, -0.18, 0.90] | apt28 ↔ executes = 0.90<br>executes ↔ powershell = 0.93<br>apt28 ↔ powershell = 0.91 |
| 8 | apt28 → [-0.02, 0.50, -2.45, -0.15, 0.92]<br>executes → [-0.01, 0.48, -2.48, -0.14, 0.94]<br>powershell → [-0.01, 0.52, -2.50, -0.12, 0.95] | apt28 ↔ executes = 0.92<br>executes ↔ powershell = 0.94<br>apt28 ↔ powershell = 0.93 |

| Layer | Embedding | Similarity |
| --- | --- | --- |
| 9 | apt28 → [0.00, 0.55, -2.55, -0.10, 0.98]<br>executes → [0.01, 0.53, -2.58, -0.09, 1.00]<br>powershell → [0.01, 0.58, -2.60, -0.08, 1.00] | apt28 ↔ executes = 0.94<br>executes ↔ powershell = 0.95<br>apt28 ↔ powershell = 0.93 |
| 10 | apt28 → [0.02, 0.60, -2.60, -0.05, 1.05]<br>executes → [0.03, 0.58, -2.63, -0.04, 1.07]<br>powershell → [0.03, 0.62, -2.65, -0.04, 1.08] | apt28 ↔ executes = 0.95<br>executes ↔ powershell = 0.96<br>apt28 ↔ powershell = 0.95 |
| 11 | apt28 → [0.04, 0.65, -2.65, -0.03, 1.10]<br>executes → [0.05, 0.63, -2.68, -0.02, 1.11]<br>powershell → [0.05, 0.68, -2.70, -0.02, 1.12] | apt28 ↔ executes = 0.96<br>executes ↔ powershell = 0.97<br>apt28 ↔ powershell = 0.96 |
| 12 | apt28 → [0.06, 0.70, -2.70, -0.01, 1.15]<br>executes → [0.07, 0.68, -2.73, 0.00, 1.16]<br>powershell → [0.07, 0.72, -2.75, 0.00, 1.18] | apt28 ↔ executes = 0.97<br>executes ↔ powershell = 0.98<br>apt28 ↔ powershell = 0.97 |

| | Layer 1 | Layer12 |
|---|---|---|
| MHA | apt28 → [-1.10, -1.05, -0.78, -0.75, -0.40]<br>executes → [-0.90, -1.40, -0.45, -0.70, -0.42]<br>powershell → [-1.02, -0.30, -0.80, -0.62, -0.70] | apt28 → [0.06, 0.70, -2.70, -0.01, 1.15]<br>executes → [0.07, 0.68, -2.73, 0.00, 1.16]<br>powershell → [0.07, 0.72, -2.75, 0.00, 1.18] |
| N1 | apt28 → [-0.95, -0.80, -0.60, -0.58, -0.20]<br>executes → [-0.70, -1.10, -0.30, -0.50, -0.25]<br>powershell → [-0.85, -0.10, -0.65, -0.45, -0.50] | apt28 → [0.05, 0.65, -2.60, -0.02, 1.10]<br>executes → [0.06, 0.64, -2.62, -0.01, 1.12]<br>powershell → [0.06, 0.67, -2.64, -0.01, 1.13] |
| FFN | apt28 → [-0.80, -0.60, -0.55, -0.50, -0.10]<br>executes → [-0.60, -0.90, -0.20, -0.45, -0.15]<br>powershell → [-0.70, 0.00, -0.60, -0.40, -0.30] | apt28 → [0.06, 0.68, -2.68, -0.01, 1.14]<br>executes → [0.07, 0.66, -2.70, 0.00, 1.15]<br>powershell → [0.07, 0.70, -2.72, 0.00, 1.17] |
| N2 | apt28 → [-0.75, -0.55, -0.50, -0.48, -0.05]<br>executes → [-0.55, -0.80, -0.15, -0.40, -0.10]<br>powershell → [-0.65, 0.05, -0.55, -0.35, -0.25] | apt28 → [0.06, 0.70, -2.70, -0.01, 1.15]<br>executes → [0.07, 0.68, -2.73, 0.00, 1.16]<br>powershell → [0.07, 0.72, -2.75, 0.00, 1.18] |
| Similarity | apt28 ↔ executes = 0.61<br>executes ↔ powershell = 0.64<br>apt28 ↔ powershell = 0.60 | apt28 ↔ executes = 0.97<br>executes ↔ powershell = 0.98<br>apt28 ↔ powershell = 0.97 |


Entity-wise Performance
1.00
0.99
0.98
0.97
0.96
0.95
0.94
0.93
0.92
Score
Precision
Recall
F1-Score
Vulnerability
Exploit
Threat-Actor-Group
Threat-Actor-Organization
Malware-Ransomware
Malware-Trojan
Malware-Worm
Attack-Initial-Access
Attack-Execution
Attack-Persistence
Attack-Privilege-Escalation
Indicator-IP
Indicator-Domain
Indicator-URL
Indicator-File-Hash
Tool
File-Name
Process-Name
Location-Country
Location-City
Time-Date
Time-Duration
Campaign
Organization
Entity Types

## Sample Sentences from Input

A threat actor known as APT29 launched a phishing campaign targeting Acme Corp and BlueRetail Ltd.
[Threat-Actor] "APT29" (tokens 5–5)
[Attack-Pattern] "phishing" (tokens 8–8)
[Organization] "Acme Corp" (tokens 11–12)
[Organization] "BlueRetail Ltd" (tokens 14–15)

The attackers sent emails containing a malicious invoice.pdf attachment that attempted to deliver the TrickBot malware.
[File] "invoice.pdf" (tokens 7–7)
[Malware] "TrickBot" (tokens 14–14)

Attackers are taking advantage of these tools to make attribution difficult and reduce the cost for developing attack infrastructure.
(no entities detected)

## Identified Entity Distribution for input

| Entity Type | Frequency |
|---|---|
| Threat Actor | 4 |
| Malware | 5 |
| Tool | 4 |
| Organization | 4 |
| File | 4 |
| Network Traffic | 3 |
| Vulnerability | 2 |
| Indicator | 2 |
| Location | 2 |
| Attack Pattern | 2 |

# Graph Construction

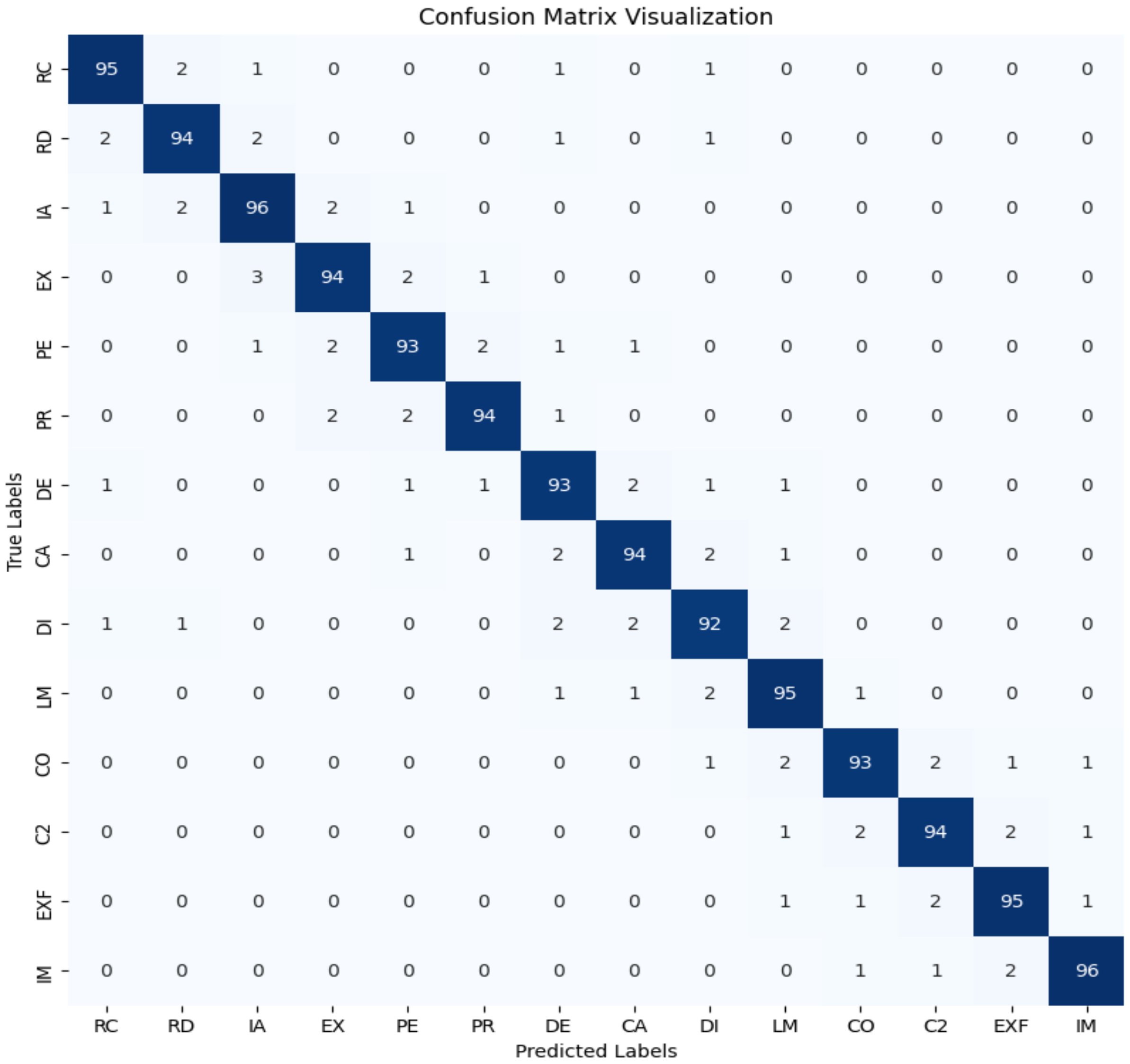
| | APT28 | exec | PS |
|---|---|---|---|
| APT28 | 0.98 | 0.55 | 0.40 |
| executes | 0.51 | 0.99 | 0.62 |
| PowerShell | 0.47 | 0.60 | 0.97 |

APT28 → executes (0.55)
APT28 → PowerShell (0.40)
executes → APT28 (0.51)
executes → PowerShell (0.62)
PowerShell → APT28 (0.47)
PowerShell → executes (0.60)

APT28 → executes
executes → PowerShell
APT28 → PowerShell

APT28 → executes → PowerShell

# Forecasting




Overall Performance

| Metric | Value |
|---|---|
| Accuracy | 95.1% |
| Macro Precision | 94.2% |
| Macro Recall | 94.0% |
| Macro F1 | 94.1% |

RC = Reconnaissance
RD = Resource Development
IA = Initial Access
EX = Execution
PE = Privilege Escalation
PR = Persistence
DE = Defense Evasion
CA = Credential Access
DI = Discovery
LM = Lateral Movement
CO = Collection
C2 = Command & Control
EXF = Exfiltration
IM = Impact

# Dynamic Causal Attack Graph Construction

**From Unstructured Cyber Threat Intelligence Reports**

## CTI Report Analysis Pipeline

Upload a CTI report or paste text

Upload CTI Report (.txt)

Drag and drop file here

Limit 200MB per file • TXT

Browse files

Or paste CTI report text here

Paste your CTI report text here...

Deploy

# 🛡️ Threat Predictor

Predict future cyber attacks

Enter the event

sen

Press Ctrl+Enter to apply

Analyze

# References

[1] Y. Wang, Z. Li, and H. Zhang, "DcBERT-IC: A Framework for Cyber Threat Analysis Integrating Deep Learning Model and Attack Intelligence Graph," *IEEE Transactions on Consumer Electronics*, vol. 71, no. 1, pp. 1–12, 2025.

[2] J. Liu, X. Chen, and Y. Wang, "MTRetrieval: Retrieving MITRE Techniques from Unstructured Cyber Threat Intelligence Reports by Fusing Deep Learning and Ontology," *IEEE Transactions on Network and Service Management*, vol. 21, no. 2, pp. 1–14, 2024.

[3] A. Rahman, K. Chen, and S. Noel, "Nip in the Bud: Forecasting Emerging Post-Exploitation Attacks in Real-Time Through CTI Reports," *IEEE Transactions on Dependable and Secure Computing*, vol. 22, no. 1, pp. 1–15, 2025.

[4] Y. Zhang, H. Liu, and J. Xu, "TRACE: Relationship Analysis and Causal Factor Extraction for Cyber Threat Intelligence Reports," *IEEE Transactions on Dependable and Secure Computing*, vol. 22, no. 2, pp. 1–13, 2025.